

# Panel data II: Difference-in-Differences

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Applied Quantitative Methods II  
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- Interpret the DiD regression coefficient
- Use event studies to visualize dynamic effects and test pre-trends
- Understand the staggered DiD problem and modern solutions

# Roadmap

From Fixed Effects to DiD

The DiD Estimator

Parallel Trends and Its Threats

Event Studies

Staggered DiD and Recent Advances

Wrap-up

## Recap: what fixed effects does

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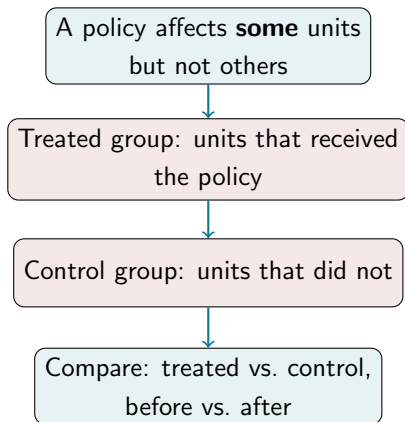
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- $\hat{\beta}$ : identified from within-unit variation, net of common trends
- But TWFE treats  $x_{it}$  as a continuous variable that varies continuously over time
- What if the variation comes from a **specific, discrete intervention**?

## A new question: policy interventions



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  - **Control:** fast-food restaurants in Pennsylvania (PA)
  - **Before:** Feb–Mar 1992; **After:** Nov–Dec 1992
- Pennsylvania did not change its minimum wage

# The 2×2 DiD table

|              | Before            | After              |
|--------------|-------------------|--------------------|
| Control (PA) | $\bar{y}_{C,pre}$ | $\bar{y}_{C,post}$ |
| Treated (NJ) | $\bar{y}_{T,pre}$ | $\bar{y}_{T,post}$ |

$$\hat{\delta}_{DiD} = (\bar{y}_{T,post} - \bar{y}_{T,pre}) - (\bar{y}_{C,post} - \bar{y}_{C,pre})$$

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- DiD subtracts this “counterfactual trend” from the treated group change

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- $\hat{\delta} = 0.59 - (-2.17) = +2.76$
- Interpretation: the minimum wage **raised** NJ employment by  $\approx 2.76$  FTEs relative to PA

# The regression formulation

$$y_{it} = \alpha + \beta_1 \underbrace{Post_t}_{\text{time}} + \beta_2 \underbrace{Treat_i}_{\text{group}} + \delta \underbrace{(Post_t \times Treat_i)}_{\text{interaction = DiD}} + \varepsilon_{it}$$

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- $\beta_2$ : pre-period difference between groups

## Regression: recovering the $2 \times 2$ cells

|                                | Before ( $Post = 0$ ) | After ( $Post = 1$ )                  |
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- The regression recovers  $\hat{\delta}$  automatically via OLS

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- Always cluster standard errors at the **treatment level**

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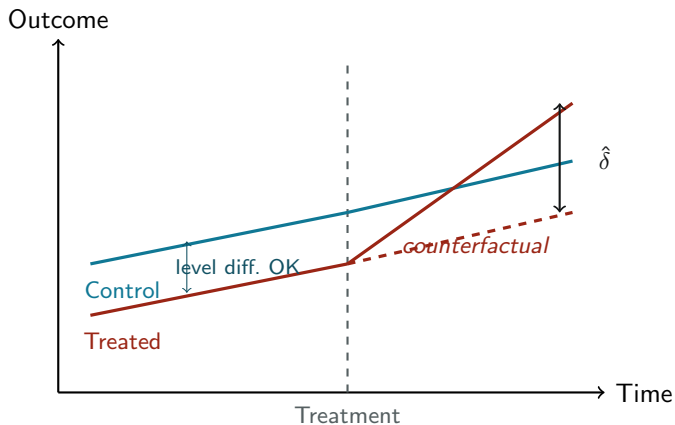
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- This is an **assumption**, not a testable fact  
(we cannot observe the counterfactual)

# Parallel trends: visualization



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- **Anticipation effects**

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- Example: firms start hiring/firing when the wage increase is announced

The parallel trends assumption says treatment would not have changed the outcome trajectory.

When is this plausible?

What makes NJ and PA a good comparison?

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- This is the **event study** design

# Event study regression

$$y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \delta_k \cdot \mathbf{1}[t - T_i^* = k] + \varepsilon_{it}$$

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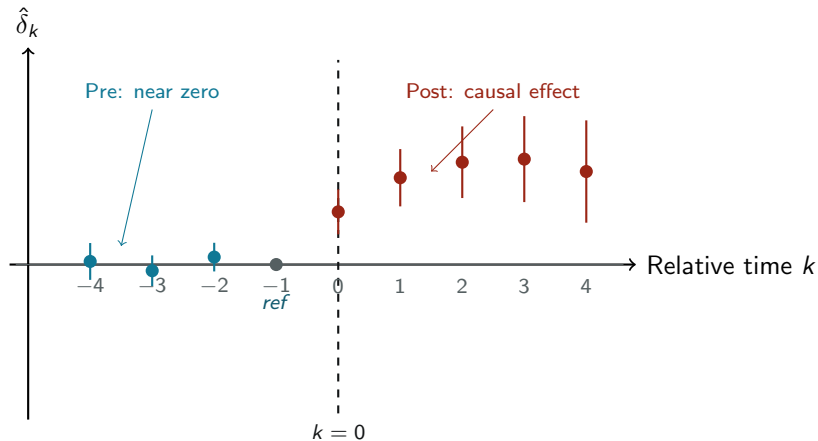
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- **Post-treatment** ( $k \geq 0$ ): trace out dynamic causal effects
- Unit and time FE absorb levels;  $\delta_k$  captures relative changes

# Event study: coefficient plot



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- `ref = -1`: omit  $k = -1$  as reference period

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- Using `feols()` with `i()` for interaction syntax:

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- `fixest` makes event studies very easy to run and plot

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- **Symmetry:** sometimes useful to check behavior far pre-treatment

# Roadmap

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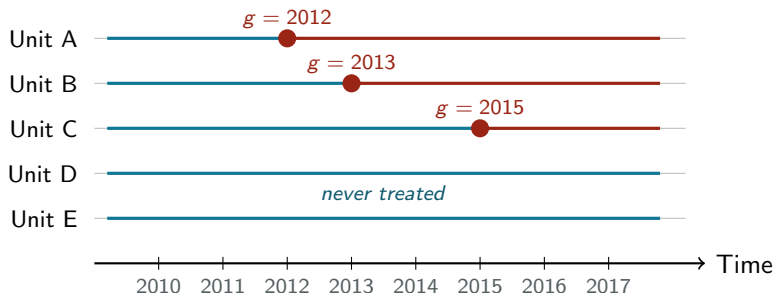
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# Staggered treatment adoption



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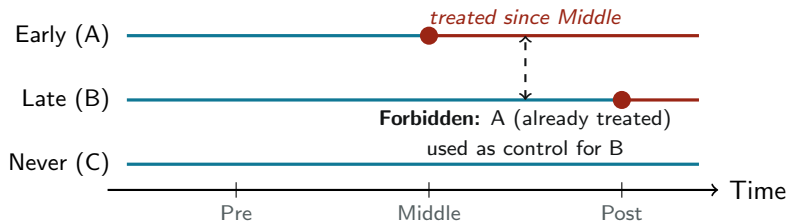
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- Result: TWFE can produce a **weighted average with negative weights**
  - Estimate may not correspond to any valid ATT

# Negative weights: the intuition



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- In R: `etwfe` package by Grant McDermott

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emfx(m) # marginal effects = ATT
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# Naive TWFE vs. modern estimators

|                       | Naive TWFE      | Callaway-Sant'Anna | ETWFE    |
|-----------------------|-----------------|--------------------|----------|
| Clean control groups  | No              | Yes                | Yes      |
| Heterogeneous effects | Biased          | Handles            | Handles  |
| Negative weights      | Possible        | None               | None     |
| Pre-trend test        | Via event study | Built-in           | Via emfx |
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# Roadmap

From Fixed Effects to DiD

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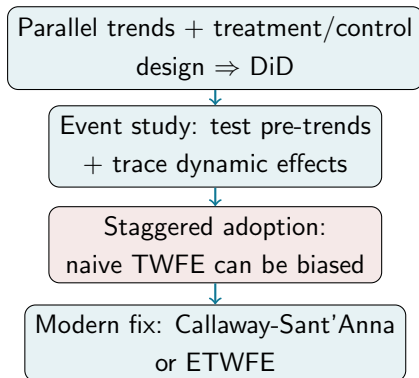
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# DiD: putting it all together



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  - Use Callaway-Sant'Anna (`did`) or ETWFE (`etwfe`)
- Always **cluster SEs** at the treatment-assignment level

# For next session

- Complete Assignment 6 (DiD application)
- Read the assigned paper (DiD design)
- Next session: Spatial Data (I)
  - Spatial data structures and visualization
  - Spatial autocorrelation
  - Spatial regression models

Questions?